

COURSE TITLE : DESCRIPTIVE ENGINEERING
COURSE CODE : 4091
COURSE CATEGORY : B
PERIODS/WEEK : 6
PERIODS/SEMESTER: 84/4 (42 +42)
CREDITS : 4 (2+2)

TIME SCHEDULE

Module	Topics	Period
1	Principles of Direct Current (DC) & Alternate current(AC) generators and motors	18
2	Electrical measuring instruments. Utilization of power and safety	24
3	Understand the different types of metals and alloys Understand various fuels	18
4	Understand the working of air compressors Understand the working of steam Boilers, their mountings and accessories, properties of steam	24
TOTAL		84

COURSE OBJECTIVE: part –A Fundamentals of Electrical Engg.

Sl.	Sub	Student will be able to
1	1	Understand the principles of Direct Current (DC) generators and DC, AC motors.
2	1	Understand electrical measuring instruments
	2	Understand the utilization of power and safety
Course Objective : <u>PART –B : Fundamentals of Mechanical engg.</u>		
3	1	Understand the different types of metals and alloys Understand various fuels
4	1	Understand the working of air compressors
	2	Understand the working of steam Boilers, their mountings and accessories Properties of steam.

MODULE I

1.1.0 Understand the principles of Direct Current (DC) generators and AC motors.

- 1.1.1 Define Faraday's law of electromagnetic induction.
- 1.1.2 Distinguish between the self and mutual induction.
- 1.1.3 State the method of obtaining alternating current.
- 1.1.4 Define: Cycle, frequency, time period, RMS value, average value, form factor, Peak factor etc
- 1.1.5 Draw the constructional features and state the working principle of DC & AC motor and generators.
- 1.1.6 Classify the various DC & AC generators.
- 1.1.7 List out applications of DC generators and alternators.
- 1.1.8 State the various methods of speed control of DC & AC motors.
- 1.1.9 State the necessity of starter.
- 1.1.10 State the working principle of 3 phase and single phase induction motor.
- 1.1.11 Classify induction motors as per motor construction.
- 1.1.12 Mention the methods of starting of 3 phase and single phase induction motor.
- 1.1.13 Mention the criteria of selection of motors for particular operations.
- 1.1.14 Classify types of drives.
- 1.1.15 List out advantages and disadvantages of electric drives.

MODULE II

2.1.0 Understand electrical measuring instruments.

- 2.1.1 Classify electrical measuring instruments.
- 2.1.2 Draw the construction and state the working principle of moving coil and moving voltmeter and ammeter.
- 2.1.3 Draw the construction and state the working principle of dynamometer type wattmeter.
- 2.1.4 Sketch and explain megger
- 2.1.5 List out applications of megger

2.2.0 Understand the utilization of power and safety

- 2.2.1 Discuss the advantages and disadvantages of electric heat
- 2.2.2 Classify types of electric heating
- 2.2.3 Give the meaning of resistance heating, induction heating, eddy current heating and dielectric heating
- 2.2.4 Discuss safety precautions from electric shock
- 2.2.5 Earthing-necessity of earthing
- 2.2.6 Methods of earthing
- 2.2.7 Earthing of 230V AC single-phase motor-circuit diagram, current path through earth.

MODULE III PART –B : FUNDAMENTALS OF MECHANICAL ENGG

3.1.0 Understand the different types of metals and alloys

- 3.1.1 Classify various metals

- 3.1.2 Explain different mechanical properties of metals
- 3.1.3 Explain different types of cast iron
- 3.1.4 Explain the manufacturing of cast iron and operation of cupola furnace
- 3.1.5 Classify different types of steels
- 3.1.6 Explain with sketches, the production of steel by electric furnaces
- 3.1.7 Explain the need for alloying steel with other elements
- 3.1.8 Describe the properties of different types of steels
- 3.1.9 List the need for the use of non-ferrous metals and alloys in engineering operations
- 3.1.10 Explain the suitability of different metals and alloys for making polymer processing machine parts and moulds.
- 3.2.0 Understand various fuels
 - 3.2.1 List the classification of Fuels and name
 - 3.2.2 List the various fuels under each classification
 - 3.2.3 Describe various fuels used in boilers, and heat exchangers.
 - 3.2.4 List the requirements of a good fuel.
 - 3.2.5 Enumerate the merits and demerits of solid, liquid and gaseous fuels
 - 3.2.6 Explain higher and lower calorific value of fuels (simple problems)

MODULE IV

4.1.0 Understand the working of air compressors

- 4.1.1 Understand the construction and working of air compressors
- 4.1.2 State the uses of compressed air.
- 4.1.3 Explain the function of an air compressor
- 4.1.4 Classify air compressor
- 4.1.5 Explain with simple sketches the working of Reciprocating compressors (single stage and multi stage)
- 4.1.6 State the advantages of internal cooling
- 4.1.7 Describe the various types of rotary compressors.

4.2.0 Understand the working of steam Boilers, their mountings and accessories

- 4.2.1 Define a boiler.
- 4.2.2 Explain the function of a boiler.
- 4.2.3 Distinguish different types of boilers.
- 4.2.4 Distinguish the fire tube boiler and water tube boiler.
- 4.2.5 Distinguish between the low-pressure boiler and high-pressure boiler.
- 4.2.6 Describe with a line diagram, the construction and working of Cochran boiler
- 4.2.7 Compare the smoke tube and water tube boilers.
- 4.2.8 Describe the package type industrial boilers.
- 4.2.9 Prepare the specifications of a boiler.
- 4.2.10 State the function of boiler, mountings such as stop – valve, safety valve, water level indicator, pressure gauge, and fusible plug with line sketches.
- 4.2.11 Explain the functions of boiler accessories such as economizer feed pump, super heater and air pre-heater, steam traps with simple sketches.

- 4.2.12 Distinguish between the boilers mountings and accessories.
- 4.2.13 Explain the draft, natural draft, induced draft and forced draft and balanced draft.
- 4.2.14 Explain the terms thermal efficiency and power of boilers
- 4.2.15 Explain the use of steam in rubber and plastic industries
- 4.2.16 Properties of steam .

CONTENT DETAILS

MODULE I

Generators and motors Alternating Current-Faraday's law of electromagnetic induction, self-induction and mutual induction Method of production of alternating current. Define the terms cycle, term periods, frequency, R.M.S value, average value and form factors, Generators: - Constructional features and working principles of DC and AC Generators. Classification of DC generators – characteristics of generators, alternators and their applications. D.C. Motors: - Principle of operation of D.C. motor, speed control of D.C. motor, flux control and voltage control strategy of D.C motors. Necessity of starter in D.C motor – application of D.C. motors AC Motors: - Principle of single phase and three phase induction motors. Classification of Induction motor base Rotor construction. Starting of 3 and single phase induction motors. Criteria for selection of motors for particular operations. Types of drives like individual, group drive, multiple drive and multi-motor drive. Advantages and disadvantages of the above drive.

MODULE II

Electric Measuring Instruments: - Classifications of electrical measuring instrument, construction and working principles of moving coil and moving iron type volt meter and ammeters, construction and working principle of dynamic meter type, watt meters. Construction and working of meggar. Application Utilization and safety Utilization: - Relays contacts, thermo stat and heating elements. Advantages and disadvantages of electric heating. Types of electric heating: Resistance heating, induction heating, Eddy current heating, dielectric heating etc. Microwave heating, IR (Infra red) heating system. Law's of electroplating – method of electroplating – methods of electroplating

Safety: - Safety precautions of electric shock – Earthing-Methods of earthings of megger. Limits switches, spark tester

Utilization power and safety Utilization: - Relays contacts, thermo stat and heating elements. Advantages and disadvantages of electric heating. Types of electric heating: Resistance heating, induction heating, Eddy current heating, dielectric heating etc. Microwave heating, IR (Infra red) heating system - Safety precautions of electric shock – Earthing -Methods of earthing.

PART –B : FUNDAMENTALS OF MECHANICAL ENGG.

MODULE III

Metals and alloys -Different type of metals and alloys – classifications – mechanical properties of metals and alloys – cast iron – manufacturing steels – production – alloying - Stainless Steel, Non ferrous metals – BIS specifications application in-polymer machinery and mould. Fuels-Classification – requirement of a good fuel – comparison of solids liquid and gaseous fuels – calorific value of fuels

MODULE IV

Air compressors –introductions – classifications working of reciprocating air compressor – multistage compressor – advantages of inter cooling Types of Rotary compressors.

Boilers-Function of boilers, classification – principle of working of fire tube boiler and water tube boiler with simple line sketches – brief explanation of the package type of industrial boiler, mountings and accessories – water level indicator, safety valve, pressure gauge, feed pump, economizer, super heater and air pre heater –steam traps- boiler – draught – brief explanation of forced draft and balanced draft – explanation of the terms thermal efficiency and power of boilers- applications in polymeric industries.

Properties of steam – steam formation under constant pressure – effect of pressure on temperature. Explanation of the properties of steam such as total heat of water, latent heat, total heat of steam, super heat, dryness fraction.

REFERENCE BOOKS

1. Elements of Heat Engines Vol. I – R.C.Patel & C.J. Karamchandani
2. Thermal Engineering Vol. I & II – M. Zakaria Baig
3. A Text book of Power plant Engineering – Dr. P.C. Sharma
4. Theory of Machines – Abdulla Sheriff

REFERENCE BOOKS

1. Electrical Technology Vol. I – B.L. Theraja
2. Electrical Technology Vol. II – B.L. Theraja
3. Electrical Power – Soni Gupta
4. Electrical and Electronics measurements – Sauhney
5. Electronic circuits – Boylestead
6. Electronics Devices and circuits – Motterhsed
7. Basic Electronics _ Floyd